

R11.

$$\frac{L_1}{R_1} + \frac{L_2}{R_2}$$

R13.

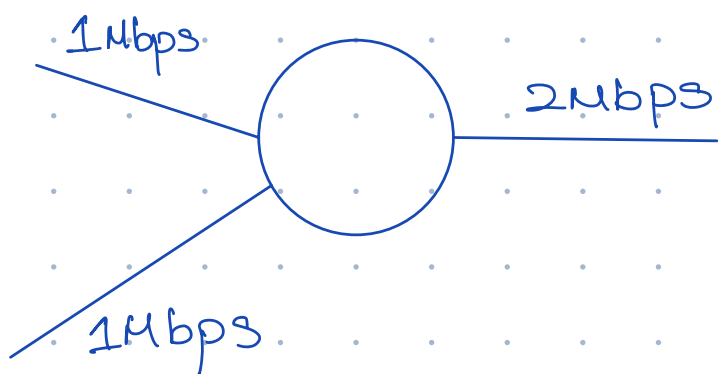
Link bandwidth = 2Mbps

Data sent = 1Mbps

Active only 20%.

$$\begin{aligned} \text{a. No. of users} &= \frac{2\text{Mbps}}{1\text{Mbps}} \\ &= 2 \end{aligned}$$

b. It is because data arrival rate would not exceed the transmission rate with 2 users but with 3 it would.



c. 0.2

d. $(0.2)^3 = 0.008$

R18.

$$\frac{2500 \times 10^3}{2.5 \times 10^8} = 10\text{ms}$$

No. & No.

R_1 .

a. $\min(R_1, R_2, R_3)$

500 kbps

b. $\frac{4 \times 10^6 \times 8}{500 \text{ kbps}} = 64 \text{ s}$

c. $R_2 = 100 \text{ kbps}$. So, this will new throughput.

Problems

P_3 .

a. Circuit switching would be a lot of data is being sent continuously, packet switching is good with bursty data.

b. No, because data arrival rate is less than link capacities, so there is no packet losses.

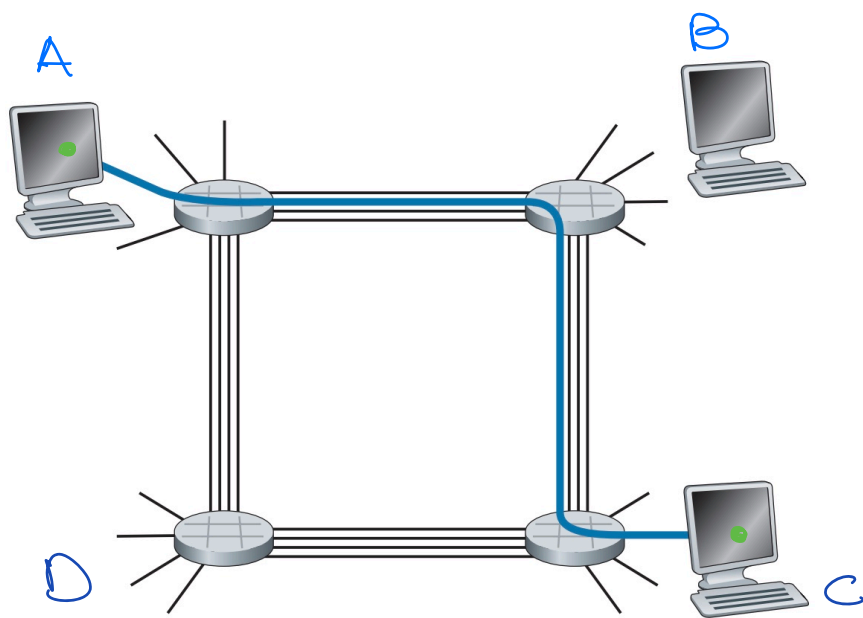
P_4 .

a. $4 \times 4 = 16$ connections

b. 4 Connections from A-B-C
& 4 from A-D-C giving
as total 8 connections
total.

c. Two from A-B-C,
Two from B-A-D
Two from A-D-C
Two from B-C-D

So, yes

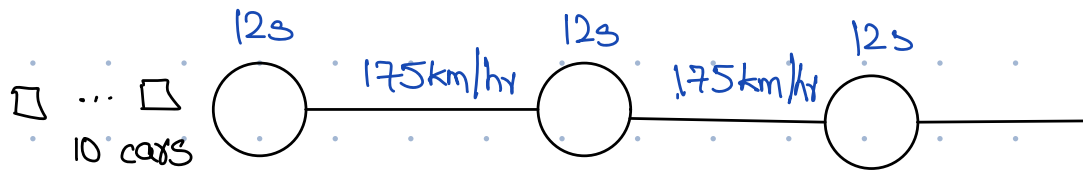


P5.

a.

$$d_{\text{prop}} = \frac{175 \text{ km}}{100 \text{ km/hr}} \\ = 1.75 \text{ hr} \Rightarrow 105 \text{ min}$$

$$d_{\text{proc}} = 10 \times 12 \\ = 120 \Rightarrow 2 \text{ min}$$



As these two roads & 3 tolls to travel. So,

$$= (105 \times 2) + (2 \times 3) \text{ min}$$

$$= 216 \text{ min}$$

P6.

$$c. \frac{L}{R} + \frac{m}{S}$$

d. Last bit will on the link

e. As $d_{\text{prop}} > d_{\text{trans}}$, so first bit is on the link

f. $d_{\text{prop}} < d_{\text{trans}}$, so first bit at dest

$$g. \frac{L}{R} = \frac{m}{S}$$

$$m = \frac{1500 \times 8}{10 \times 10^6} \times 2.8 \times 10^8 \\ = 300 \text{ km}$$

P₇.

$$\text{Packet delay} = 56 \times 8 \text{ b} \times \frac{1}{64 \times 10^3 \frac{\text{b}}{\text{s}}}$$

$$= 7 \times 10^{-3} \text{ s}$$

$$d_{\text{trans}} = \frac{56 \times 8}{10 \times 10^6}$$

$$= 4.48 \times 10^{-5} \text{ s}$$

$$= 7 \times 10^{-3} + 4.48 \times 10^{-5} + 10 \times 10^{-3} \text{ s}$$

$$= 0.017 \text{ sec}$$

P₈.

$$\text{a. No. of users} = \frac{10 \times 10^6 \text{ b/s}}{200 \times 10^3 \text{ b/s}}$$

$$= 500$$

$$\text{b. } 0.1$$

$$\text{c. } P(X=r) = {}^{120}C_r (0.1)^r \times (0.9)^{120-r}$$

$$\text{d. } P(X \geq 51) = 1 - P(X < 51)$$

$$= 1 - \sum_{i=0}^{50} {}^{120}C_i (0.1)^i \times (0.9)^{120-i}$$

P₉.

$$\text{a. no. of users} = \frac{1 \times 10^9}{1 \times 10^5}$$

$$= 10^4$$

$$\text{b. } P(X > N) = 1 - P(X \leq N)$$

P₁₀.



$$s = 2.5 \times 10^8 \text{ m/s} \quad R = 2.5 \text{ Mbps} \quad L = 1500 \text{ bytes}$$

$$d_{\text{proc}} = 3 \text{ ms}$$

$$= \frac{5000 \times 10^3}{2.5 \times 10^8} + \frac{4000 \times 10^3}{2.5 \times 10^8} + \frac{1000 \times 10^3}{2.5 \times 10^8} + 3 \left(\frac{1500 \times 8}{2.5 \times 10^6} \right) + 2(3 \times 10^{-3})$$

$$= 0.0604 \text{ sec}$$

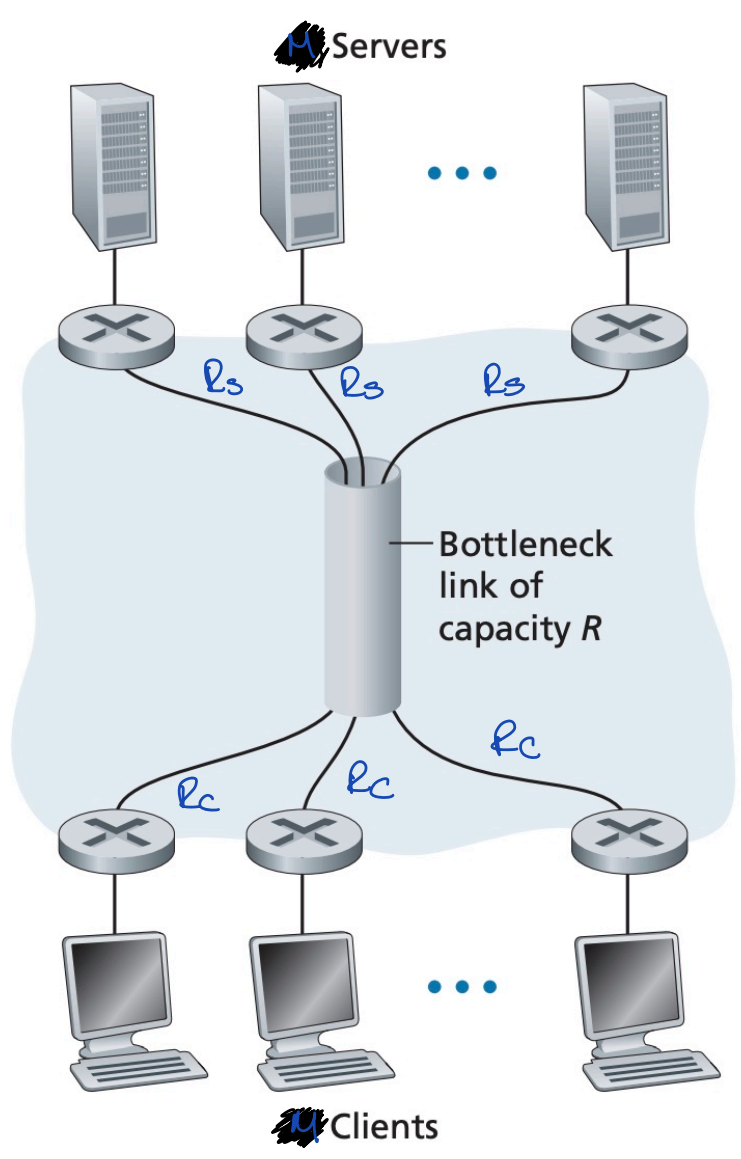
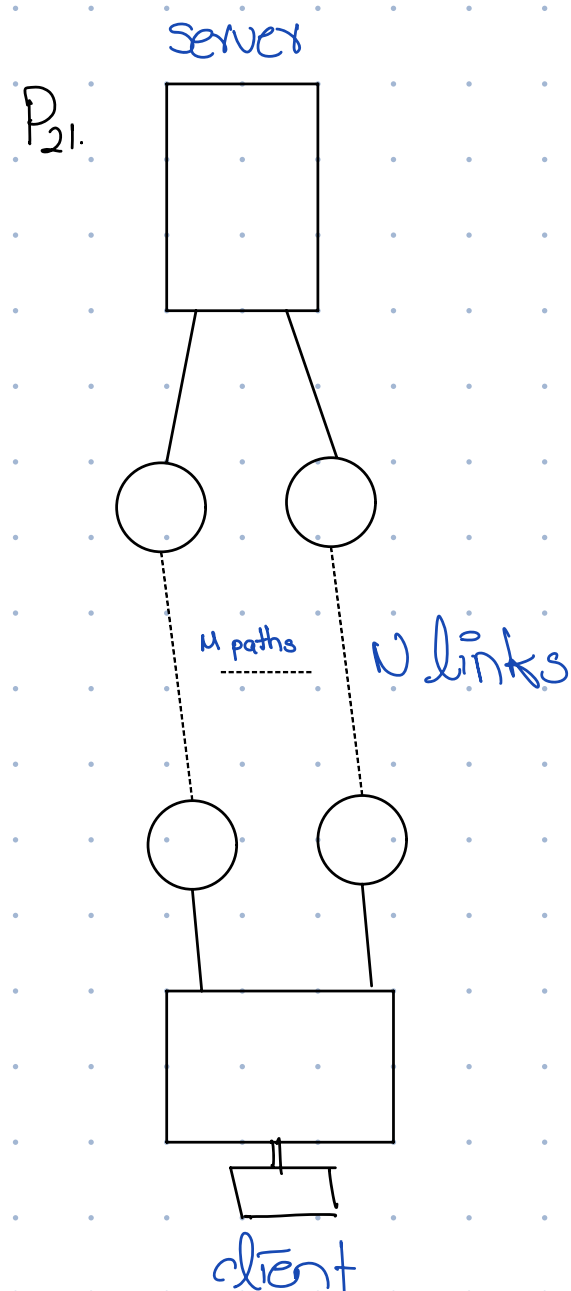
P₁₁.

$$\frac{5000 \times 10^3}{2.5 \times 10^8} + \frac{4000 \times 10^3}{2.5 \times 10^8} + \frac{1000 \times 10^3}{2.5 \times 10^8} + \underbrace{\frac{1500 \times 8}{2.5 \times 10^6}}_{\text{sender's delay}}$$

$$0.0448 \text{ sec}$$

P₂₀.

$$\min(R_s, R_c, R/M)$$



b.

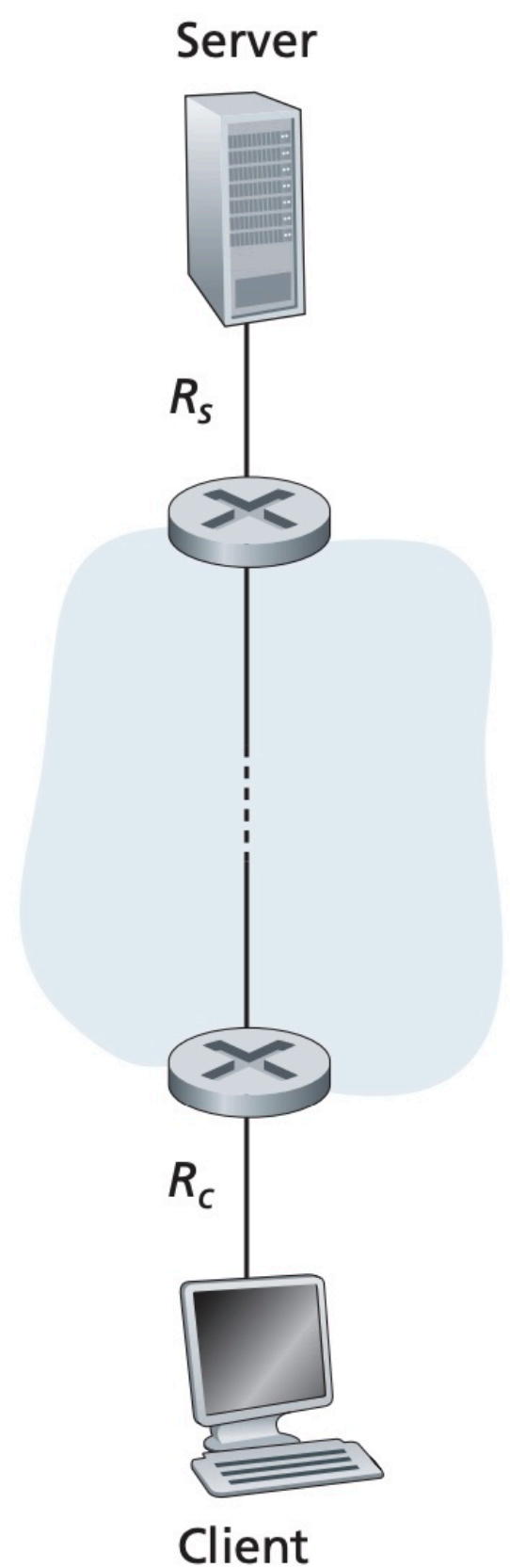
$$\max(\min(R_1, \dots, R_k)_1, \min(R_1, \dots, R_k)_2, \dots, \min(R_1, \dots, R_k)_m)$$

P22.

$$\text{Probability of packet not lost} = (1-p)^m$$

$$\text{Avg re-transmissions} = \frac{1}{(1-p)^m} - 1 \quad \text{geometric distribution}$$

P23.



a.

P₂₄.

$$= \frac{50 \times 10^{12} \times 8}{10 \times 10^6}$$

$$= 4 \times 10^7 \text{ s}$$

Fedx is better

P₂₅.

$$b. \frac{s}{2.5 \times 10^8 \text{ m}} \times 20,000 \times 10^3 \times 5 \times 10^6 \frac{\text{b}}{\text{s}}$$

4000000 bits

P₂₈

$$b. \left[\frac{40,000}{5 \times 10^6} + \frac{20,000 \times 10^3}{2.5 \times 10^8} \times 2 \right] 20$$

$3 + 4 = 7$

3.36 s

$$a. \frac{800,000}{5 \times 10^6} + \frac{20,000 \times 10^3}{2.5 \times 10^8}$$

0.32 s

c. (a) is faster due to lack of packet delay

P₂₉

$$a. \frac{36,000 \times 10^3}{2.4 \times 10^8} = 0.15 \text{ s}$$

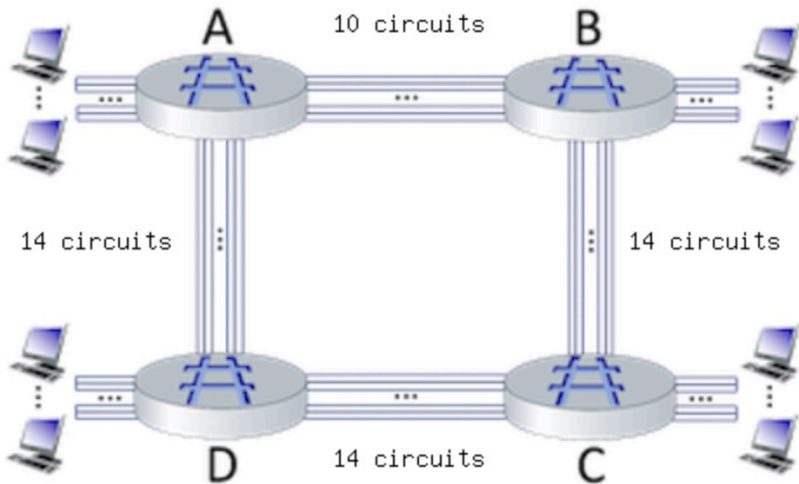
c. $60 = \frac{2}{10 \times 10^6}$

$2 = 600,000,000 \text{ bits}$

Circuit Switching

CIRCUIT SWITCHING

Consider the circuit-switched network shown in the figure below, with circuit switches A, B, C, and D. Suppose there are 10 circuits between A and B, 14 circuits between B and C, 14 circuits between C and D, and 14 circuits between D and A.



QUESTION 1 OF 4

What is the maximum number of connections that can be ongoing in the network at any one time?

$10 + 14 + 14 + 14$
 52

QUESTION 2 OF 4

Suppose that these maximum number of connections are all ongoing. What happens when another call connection request arrives to the network, will it be accepted? Answer Yes or No

No

QUESTION 3 OF 4

Suppose that every connection requires 2 consecutive hops, and calls are connected clockwise. For example, a connection can go from A to C, from B to D, from C to A, and from D to B. With these constraints, what is the maximum number of connections that can be ongoing in the network at any one time?

A - B, B - C
 B - C, C - D
C - D, D - A
D - A, A - B

Find hops common links & sum them & then take min & also opposite ones.

$$- 10 + 14 = 24$$

$$- 14 + 14 = 28$$

So, 24